

Measuring the Effect of Iran War Risk on Global Financial Markets

FRE-GY 7871 A · NLP and the Investment Process · Fall 2026

Key takeaways

- A war-risk shock large enough to move oil up \$5 a barrel knocks about 0.65% off the S&P 500 — but 1.4-1.7% off European, Japanese and emerging-market equities. The United States is now insulated relative to the rest of the world in a way it was not in 2003, because it is a much larger oil producer.
- Credit widens, volatility rises, and the dollar and Swiss franc both strengthen — a conventional flight-to-safety pattern in currencies and credit, layered on top of an oil shock that behaves very differently from the 2003 Iraq-war episode.
- The critical break from 2003: Treasury yields and inflation expectations **rise** with war risk in 2026, where they fell in 2003. This is the signature of a supply-side shock (more oil-dependent inflation) rather than a demand-side scare, and it lines up with the structural shift in US energy production since then.
- The identification strategy was tested against the most obvious simpler alternative — a plain regression on "was this a big war-news day, yes or no" — and that alternative found nothing. Every one of eighteen markets came back statistically indistinguishable from zero. The variance-based approach used here found reliable effects in thirteen of seventeen. That gap is the strongest evidence in this analysis that the more involved method is earning its keep, not just adding complexity.
- A few readings should be treated with real caution rather than taken at face value: gold, Tel Aviv equities, and defence stocks all rest on a weaker piece of the statistical machinery and move around noticeably depending on exactly how the analysis is run. Everything else — equities, credit, oil, the VIX, the dollar, Treasury yields — holds up under several different ways of stress-testing it.

Why war risk needs a different kind of measurement

Nobody publishes a daily "probability of war" number, so there is no series to put directly into a regression. What can be observed is *when* war-related news was unusually intense. Rigobon and Sack's 2003 study of the Iraq war turned that limitation into a method: split trading days into a group where war-related news was heavy and a matched group where it was quiet, and see how much more markets moved around — not on average, but in dispersion — in the heavy-news group. If every other driver of prices was no more volatile on those days than usual, the extra dispersion is attributable to the war-risk factor alone, and its size can be recovered without ever assigning a number to "risk" itself.

The appeal for this kind of analysis is that it does not require knowing whether a given day's news was good or bad for markets — only that it was voluminous and uncertain. That matters a great deal here: of the eighteen days flagged as unusually newsy over the sample, six carried heavy coverage that was genuinely split between escalation and de-escalation at once. A method that needed a signed input would have to throw those days out or guess at their direction. This one does not.

Building the signal from the news

Every trading day between January and mid-September 2026 gets a single flag — high war-news day or not — built from GDELT's coverage of Iran-related reporting: how much coverage there was relative to its own recent baseline, how far the balance of escalation-versus-de-escalation language moved, and how contested that balance was on a

given day. Eighteen of 173 trading days clear the bar, roughly the top decile, comparable in scale to the seventeen days Rigobon and Sack identified by hand for Iraq. A matched set of eighteen quiet days, chosen to sit as close in time as possible without themselves being newsy, anchors the comparison.

Sourcing real reporting to describe what happened on those eighteen days turned out to be harder than expected. GDELT's own article-level search does index real outlets — the underlying feed carries publisher domains like reuters.com and aljazeera.com — but a systematic attempt to pull headlines through it, tested at several request speeds with generous retry logic, could not reliably get through its rate limiting within a session. Rather than fall back on a tertiary source like an encyclopedia's own event summary, each of the eighteen days below was checked by hand against dated Al Jazeera, CNN and Bloomberg reporting, with a source link kept for every entry. That verification runs after the day-selection is already finished — the flagging itself never sees this text — so it changes only what gets printed next to each date, not which days were chosen or any of the results that follow.

The eighteen days that defined the war-risk narrative

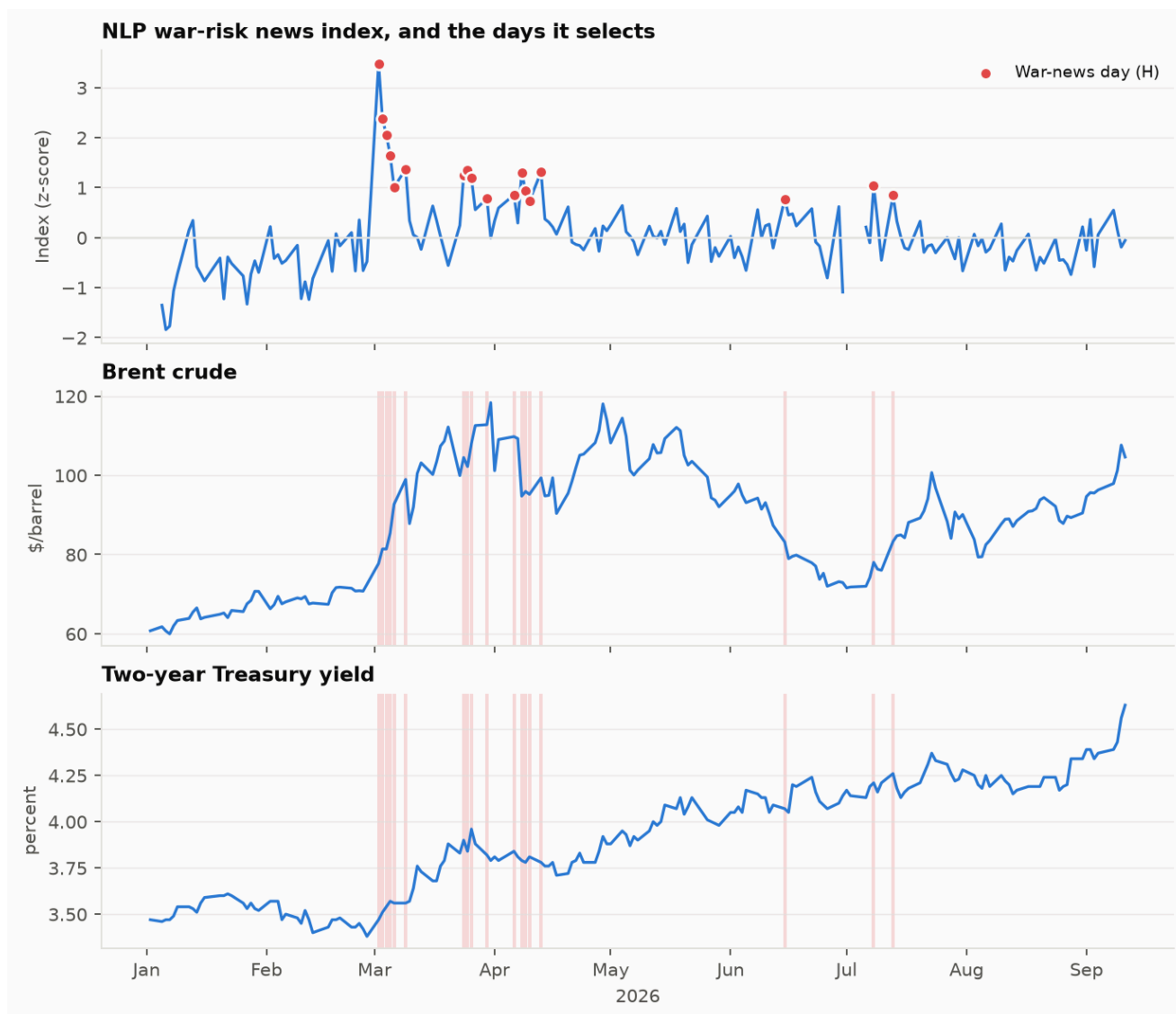
Table 1. War-news days selected by the news index (18 of 173 trading days, top decile; the Brent and two-year-yield columns are shown for context and were not used to pick the days)

Date	War risk	News score	Δ Brent (\$)	Δ 2y (bp)	Source	Event
2026-03-02	Increased	3.48	+5.26	+9.0	CNN/Al Jazeera	US/Israel strike Natanz enrichment complex; Iran closes Strait of Hormuz
2026-03-03	Increased	2.38	+3.66	+4.0	Al Jazeera	Massive explosions in Tehran, Karaj, Isfahan; IRGC says ground forces engaged
2026-03-04	Increased	2.06	+0.00	+3.0	CNN	US says Iran's air force/navy "knocked out"; >1,000 killed in Iran
2026-03-05	Increased	1.65	+4.01	+3.0	CNN	Iran fires drone/missile barrage at Tel Aviv; Brent +24% on the week
2026-03-06	Increased	1.01	+7.28	−1.0	Al Jazeera	US warns Iran hit "very hard"; >3,000 targets struck in past week
2026-03-09	Increased	1.37	+6.27	0.0	Al Jazeera	Iran names Mojtaba Khamenei Supreme Leader; Brent hits \$119.50
2026-03-24	Unclear	1.24	+4.55	+7.0	Al Jazeera/CNN	Talks claimed underway; Tehran calls it "fake news"
2026-03-25	Unclear	1.35	−2.27	−6.0	Al Jazeera	Oil tumbles, stocks rally on peace-plan reports; Iran rejects it
2026-03-26	Unclear	1.19	+5.79	+12.0	CNN	Energy-site strikes held off 10 more days as "talks ongoing"; worst month for US stocks
2026-03-30	Increased	0.78	+0.21	−6.0	Al Jazeera	Threat to obliterate energy infrastructure if no deal reached
2026-04-06	Increased	0.86	+0.74	+5.0	CNN	Deadline set to reopen Hormuz; 45-day ceasefire proposal rejected
2026-04-08	Unclear	1.31	−14.52	−2.0	Al Jazeera/CNN	Pakistan-brokered 2-week ceasefire; Iran agrees to reopen Hormuz
2026-04-09	Decreased	0.94	+1.17	−1.0	Al Jazeera	IRGC claims Hormuz shipping stopped again; fire at Aramco's Abqaiq facility
2026-04-10	Decreased	0.73	−0.72	+3.0	CNN	Hormuz to reopen "soon, with or without" Iran; polls doubt the win
2026-04-13	Decreased	1.31	+4.16	−3.0	Al Jazeera/CNN	Blockade of all Iranian ports begins after talks collapse

Date	War risk	News score	Δ Brent (\$)	Δ 2y (bp)	Source	Event
2026-06-15	Decreased	0.77	-4.16	-2.0	Bloomberg	Oil sinks >5% on interim deal to reopen Hormuz
2026-07-08	Unclear	1.04	+3.86	+2.0	Al Jazeera/CNN	Iran warns of "crushing response"; strikes 85 US targets in Bahrain/Kuwait
2026-07-13	Unclear	0.85	+7.29	+5.0	CNN/Al Jazeera	Strikes resume 2nd night; Iran disables 2 tankers; WTI +9.4%

Eight of the eighteen days lean escalatory, four lean toward de-escalation, and six carry heavy coverage that is genuinely split — the sort of day a conventional event study cannot use. The list is not evenly spread across the sample: nearly half of it falls in March alone, the month of the most intense fighting, with a second cluster around the April ceasefire and two isolated flare-ups later in the summer when the truce broke down over shipping through the Strait of Hormuz.

Nothing about the actual timeline of the war was given to the scoring process — it only ever sees coverage volume and the balance of escalation-versus-de-escalation language. Plotting the resulting index against the level of Brent crude and the two-year Treasury yield over the same window is therefore a useful sanity check rather than a formality: the index's single sharpest spike lands on February 28, the day the war began, and it climbs from a quiet, slightly negative baseline through January and February into a dense, elevated cluster through March before tapering off after the April ceasefire. Brent's price level climbs steadily across that same window; the two-year yield moves in a visibly choppier, less obviously war-linked pattern — an early clue, well before the formal identification test below, that oil is carrying this episode's risk signal far more cleanly than the Treasury market is.



What a war-risk shock does to markets, asset by asset

The eighteen flagged days are matched against the eighteen quiet comparison days, and the difference in how much each market moved translates into a sensitivity estimate, scaled here to a war-risk increase large enough to push Brent up \$5 a barrel — the same instrument Rigobon and Sack used for the two-year yield in 2003, substituted here because the two-year yield turns out not to respond distinctly enough to Iran-specific risk to serve that role in 2026 (more on that below).

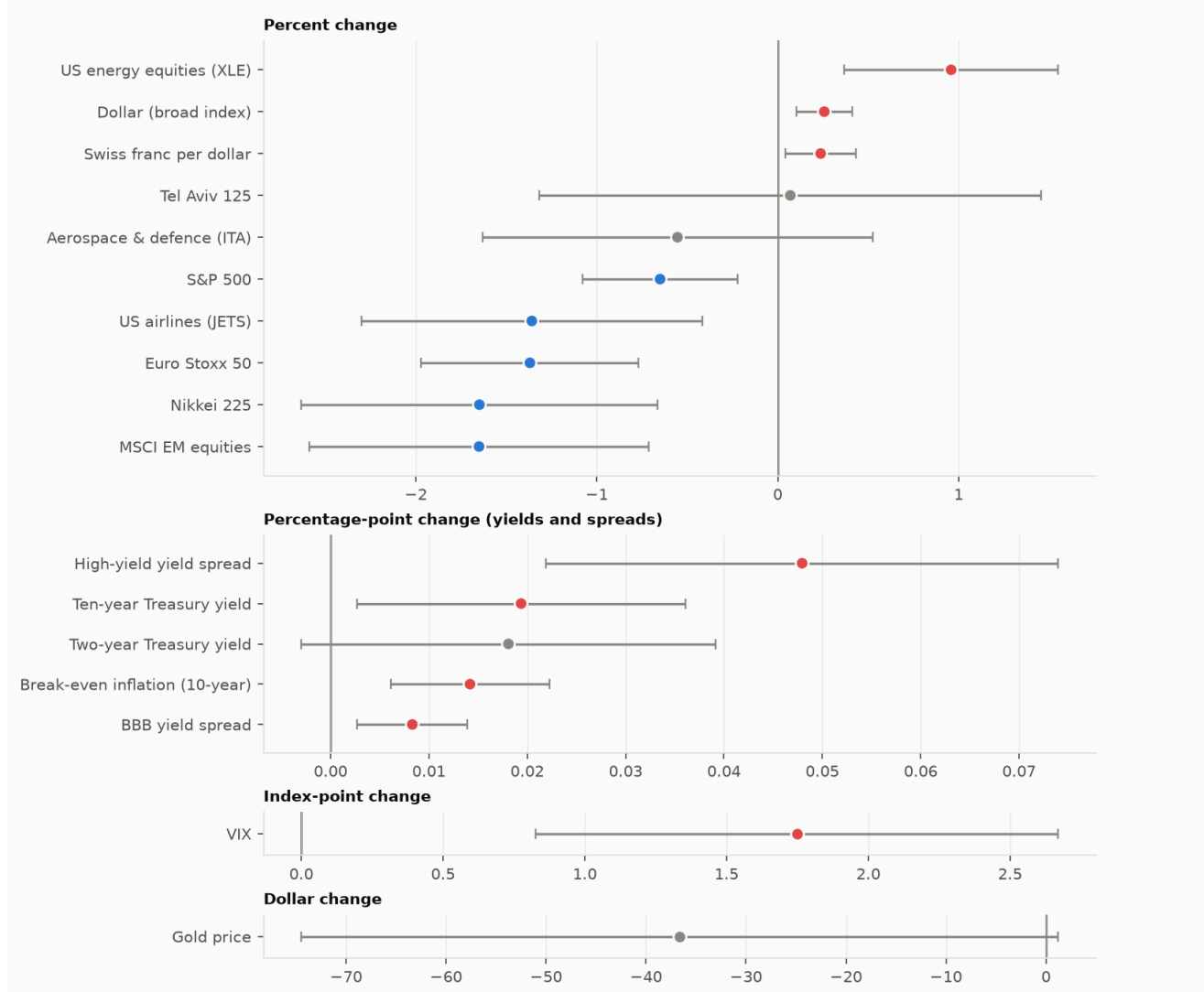
Table 2. Estimated market response to a war-risk shock that raises Brent \$5/bbl

Variable	Units	Coefficient	t-stat	Sig.
Two-year Treasury yield	pp	0.018	1.68	*
Ten-year Treasury yield	pp	0.019	2.28	**
Break-even inflation (10y)	pp	0.014	3.45	***
S&P 500	pct	-0.650	-2.96	***
BBB yield spread	pp	0.008	2.91	***

Variable	Units	Coefficient	t-stat	Sig.
High-yield yield spread	pp	0.048	3.61	***
Gold price	\$	-36.63	-1.90	*
Dollar (broad index)	pct	0.257	3.24	***
VIX	pts	1.748	3.72	***
Euro Stoxx 50	pct	-1.370	-4.47	***
Nikkei 225	pct	-1.649	-3.28	***
Tel Aviv 125	pct	0.069	0.10	
MSCI EM equities	pct	-1.651	-3.45	***
Swiss franc per dollar	pct	0.237	2.36	**
US energy equities (XLE)	pct	0.958	3.17	***
US airlines (JETS)	pct	-1.359	-2.83	***
Aerospace & defence (ITA)	pct	-0.554	-1.01	

Thirteen of seventeen coefficients clear conventional significance at 5% — plotting the same numbers as point estimates with confidence bands makes the shape of the result easier to take in at once than the table does.

Figure 3. Estimated impact of higher Iran war risk
Response to a war-risk increase that raises Brent \$5/bbl. Combined instrument, 95% intervals;
grey marks estimates not significant at 5%.



Equities fall everywhere, and fall roughly twice as hard outside the United States. The S&P gives back 0.65% for a \$5 oil shock; Europe, Japan and emerging markets all lose 1.4-1.7%, more than double the US move. That gap is the clearest single number in this analysis, and it is a structural story rather than a sentiment one: the United States pumps roughly 14 million barrels of oil a day today, against roughly 6 million in 2003, so a war-driven spike in oil prices is now a mixed blessing for the US economy rather than a uniformly negative one. That shows up directly inside the US market too — energy stocks are the only positive line in the whole table, up 0.96%, while airlines, which eat the same oil-price increase on the cost side, lose 1.36%.

Credit widens in proportion to risk, not uniformly. High-yield spreads move six times as far as investment-grade spreads (4.8 basis points versus 0.8), which is what a genuine risk repricing should look like rather than a blanket move in credit markets. The VIX adds close to 1.75 points, a real but not extreme jump in implied volatility for a shock of this size.

The break from the 2003 playbook shows up in rates, inflation, and the dollar. In the run-up to the Iraq war, all three of those fell together as investors priced war risk as a threat to growth: money moved into safe government debt, inflation expectations eased, and the dollar weakened. Here, all three do the opposite — the ten-year yield rises, break-even inflation rises, and the dollar strengthens (with the Swiss franc rising alongside it, so this reads as

a broad flight to currencies seen as safe rather than a dollar-specific story). The read is that 2026's war risk is being priced primarily as a threat to oil *supply* — inflationary, and, given the scale of the domestic energy sector, less of a pure flight-to-safety trigger for the dollar than it was in 2003.

Two results are worth flagging as non-findings rather than findings. Gold shows no reliable response (its point estimate is even the wrong sign, and its confidence interval comfortably spans zero) — which echoes Rigobon and Sack's own 2003 result and is a useful corrective to the assumption that gold reliably behaves as a war hedge. Tel Aviv equities also show no statistically reliable response, despite moving around noticeably more than usual on the flagged days — a distinction worth sitting with for a moment, because it recurs below: being volatile on newsy days and being *driven by the shared war-risk factor other assets share* are not the same thing, and Tel Aviv is the cleanest example of a market where the first is true and the second is not.

How much of the story is war risk, and how much is everything else

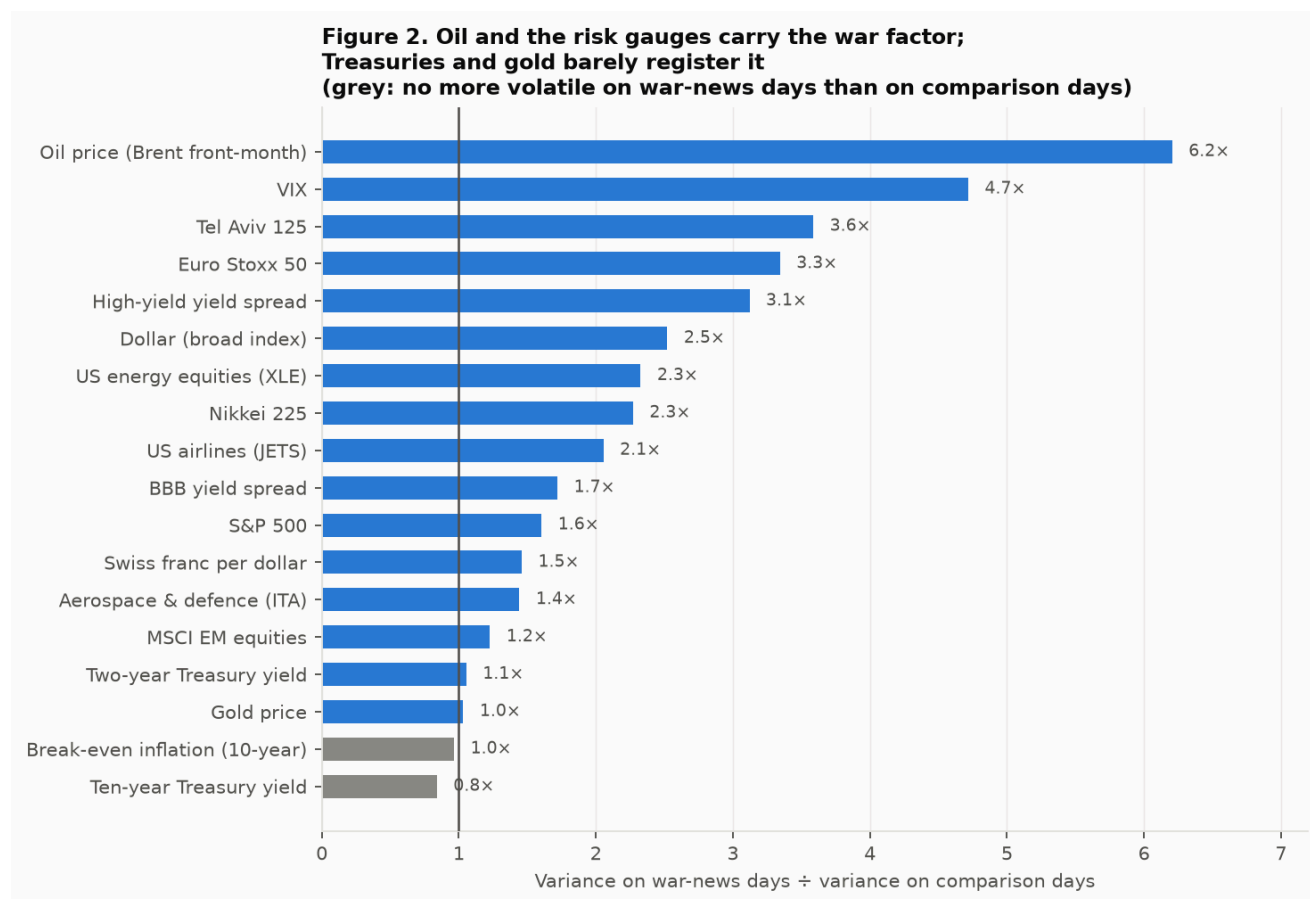
Not every point of volatility on a newsy day is attributable to the war. The share that is can be estimated by comparing how much extra variance each market carries on the eighteen flagged days against how much of that extra variance the shared war-risk factor alone can explain.

Table 3. Share of variance attributable to the war-risk factor

Variable	Share of variance, war-news days	Share of variance, full sample
Euro Stoxx 50	53.0%	16.8%
VIX	49.4%	12.5%
Dollar (broad)	43.9%	8.5%
MSCI EM equities	42.9%	8.7%
Break-even inflation	42.2%	4.6%
High-yield spread	41.3%	9.0%
S&P 500	38.7%	6.1%
Nikkei 225	33.4%	8.2%
US airlines	31.5%	3.9%
US energy equities	30.8%	4.3%
BBB spread	28.3%	5.1%
Swiss franc	20.4%	2.4%
Ten-year Treasury yield	16.8%	2.0%
Gold price	15.9%	1.6%
Two-year Treasury yield	12.5%	1.4%
Aerospace & defence	8.0%	1.3%
Tel Aviv 125	0.1%	0.0%

On the flagged days themselves, war risk accounts for roughly half of everything moving European equities and the VIX, and a sizeable chunk of the dollar, emerging markets and credit spreads. Stretched over the full eight-month sample, where most days are ordinary trading days, those shares compress to a more modest but still material 8-17% for the same markets — a reasonable order of magnitude for a single geopolitical factor operating alongside everything else that normally moves prices. These figures should be read as floors rather than precise numbers: the quiet comparison days are not perfectly war-news-free in a conflict that ran for six months, which understates the true share attributable to the war.

The underlying variance ratios behind this table are worth looking at directly, because they explain a methodological choice made earlier in this analysis: oil's variance is 6.2 times higher on war-news days than on quiet ones, the VIX's 4.7 times, and Tel Aviv's 3.6 times — but the two-year Treasury yield comes in at just 1.05 times, and the ten-year actually shows *less* variance on war-news days than on ordinary ones. That is the concrete reason Brent, not the two-year yield, anchors the estimates above: in the 2003 study the two-year yield was the cleanest available barometer of war risk, with variance rising more than sixfold on war-news days, but that role has effectively moved to oil in this episode.



The Tel Aviv result from the previous section becomes clearer against this backdrop. Its variance ratio (3.6x) is the third-highest of any market in the sample — genuinely much noisier on war-news days — yet the share of that noise attributable to the shared factor is essentially zero. Put together, the two numbers say that Israeli equities move a great deal on days when the war escalates, but for reasons specific to Israel (local political and economic news tied directly to the fighting) rather than through the same oil-and-risk-sentiment channel that moves the S&P, European equities, and credit spreads together.

Stress-testing the approach

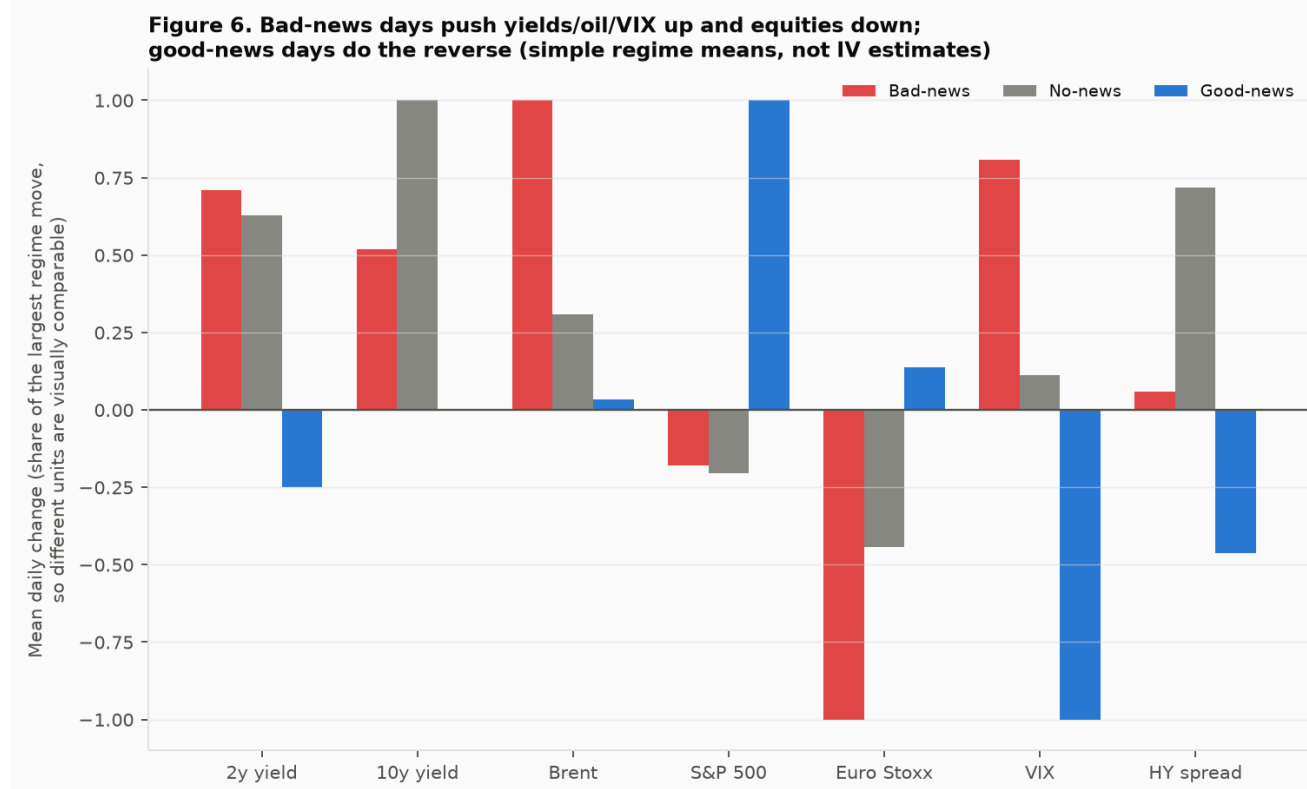
A method this specific to the problem is worth checking against simpler alternatives before trusting its output. Three checks were run.

The most obvious alternative — a plain regression of each market's daily move on the same high-news-day flag — comes back empty. Run as a straightforward test of "is the average move different on flagged days versus not," it finds zero of eighteen markets significant at the 5% level. The variance-based approach, on the identical eighteen days, finds thirteen of seventeen. The reason for the gap is not subtle once you look at the eighteen days themselves: eight lean escalatory and four lean toward de-escalation, so a market that genuinely moves in opposite

directions on bad-news versus good-news days has those moves cancel out in a simple average, pushing the estimated effect toward zero regardless of how real the underlying sensitivity is. Variance does not cancel the same way — a \$5 up-move and a \$5 down-move both add to dispersion — so the shared factor survives averaging even when its sign flips from day to day. This is the single clearest piece of evidence that the more involved method earns its complexity here rather than adding it for its own sake.

Splitting the flagged days by direction confirms the intuitive story with nothing more than simple averages.

Separating the eighteen days into those where coverage leaned escalatory, those where it leaned toward de-escalation, and comparing both against the quiet days, seven markets checked — two-year and ten-year yields, oil, the S&P, Euro Stoxx, the VIX, and high-yield spreads — all move in the expected direction on bad-news days (yields, oil and volatility up, equities down) and mostly reverse on good-news days.

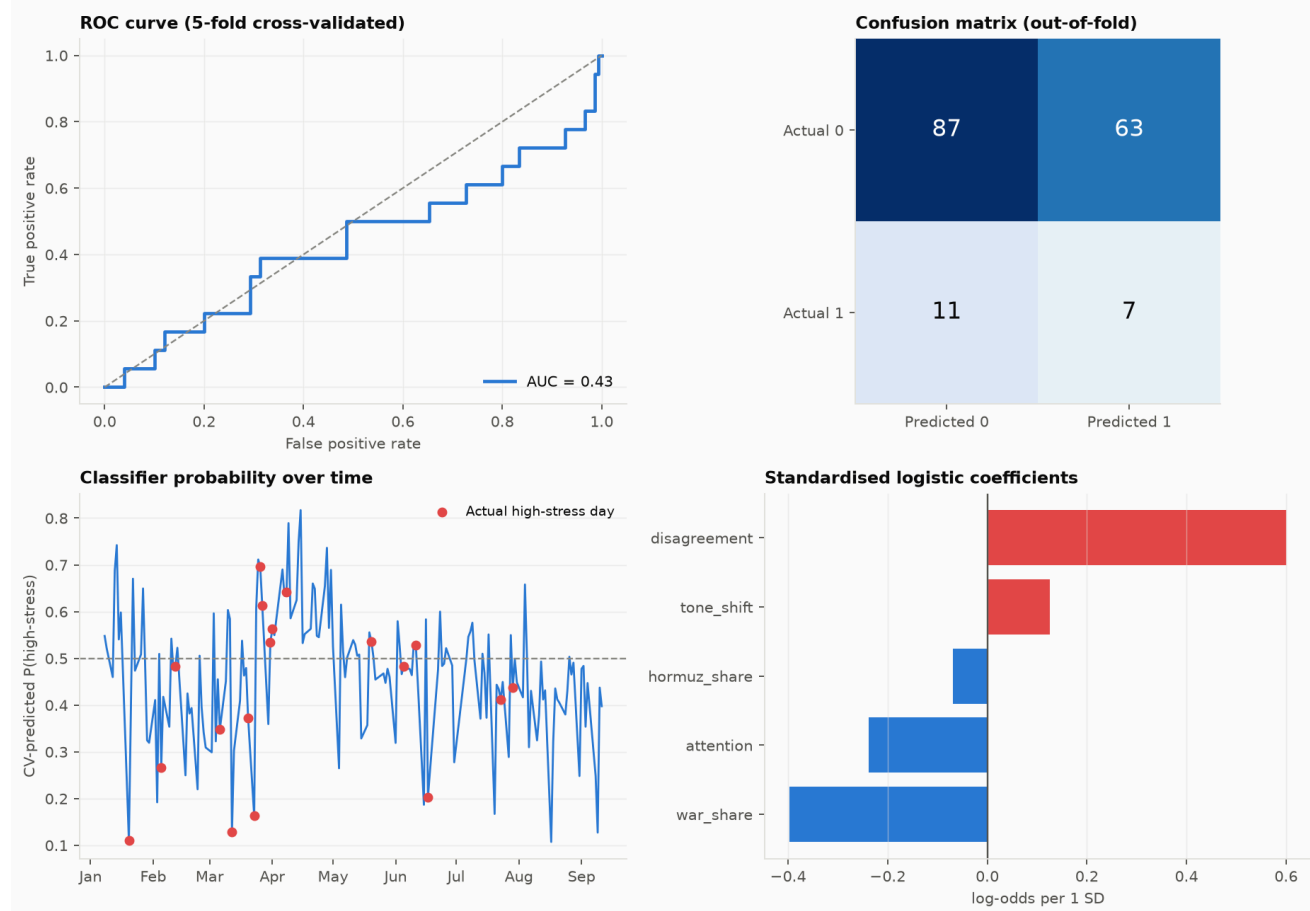


A stricter version of this check — estimating the sensitivity separately from the bad-news days and separately from the good-news days, and asking whether the two estimates agree in sign despite pulling from sub-samples with opposite average news direction — agrees on ten of seventeen markets. That is a noisier result, unsurprising given the sub-samples run to only four and eight days each, but it belongs in the record alongside the cleaner seven-for-seven finding above rather than being left out because it is less tidy.

Letting a model, rather than a fixed rule, decide which days count as newsy produces a genuinely mixed result — and the reason why is itself informative. A cross-validated classifier was trained to predict an independent, market-based definition of a high-stress day using only the news features (no price data was used in training). Its accuracy came back worse than a coin flip. Digging into why turns out to reinforce the main approach rather than undercut it: the market-based label used to train the classifier was built from a broad basket of assets that includes the two-year yield — and the two-year yield, as established above, barely responds to Iran-specific risk at all. A label partly built from a market that doesn't respond to this particular risk factor will naturally be dominated by other, unrelated sources of volatility (Fed decisions, data releases, and so on), and on those days Iran-specific news coverage is, unsurprisingly, not elevated. Only 3 of the 18 days flagged by news content overlap with the 18 days flagged by market stress — which is really a demonstration of why selecting days from the news

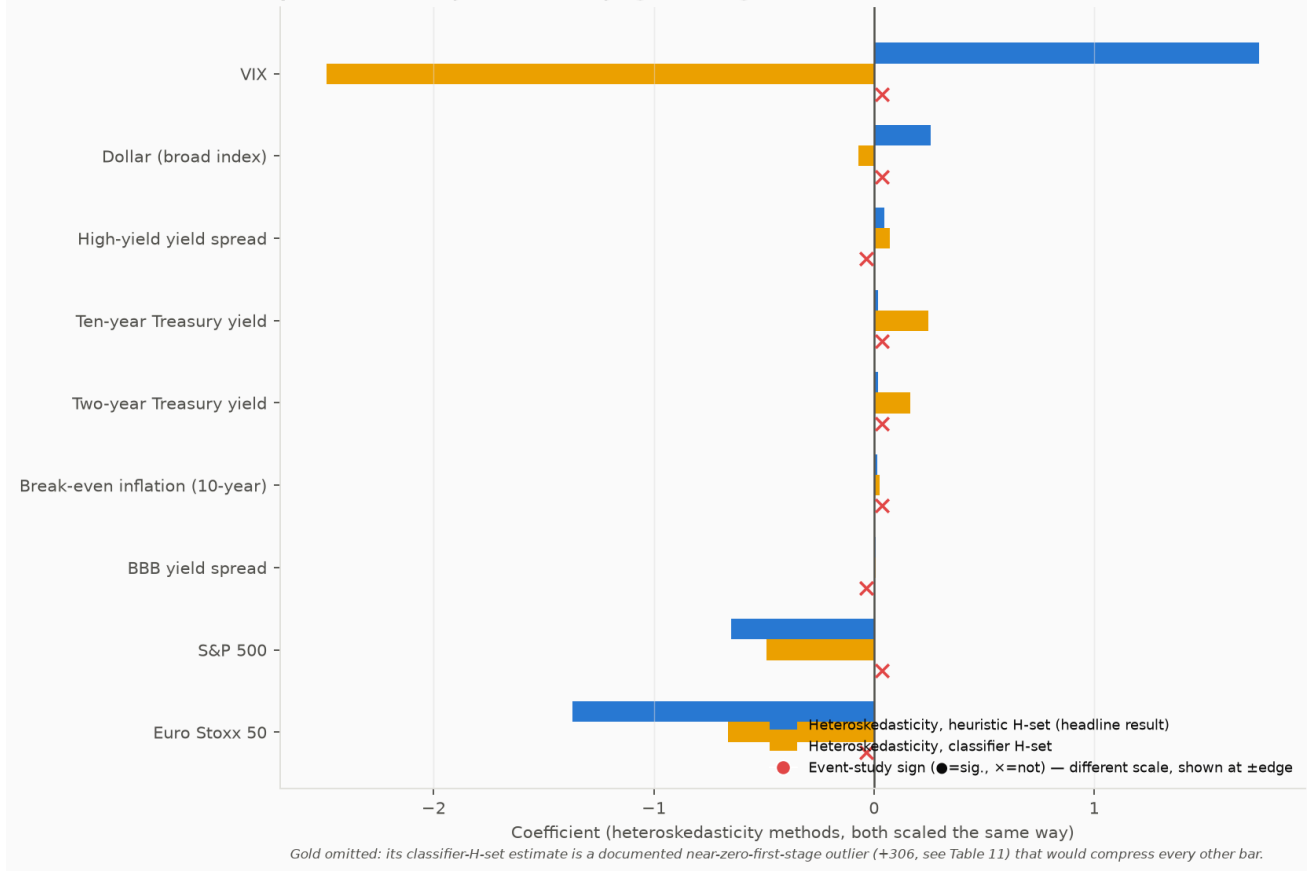
itself, rather than from realized price moves, is the right design for isolating this specific risk: a price-based selection rule would have been contaminated by everything else that moves markets, not just Iran.

Figure 4. Can NLP features predict which days will be high-market-stress days?



Re-running the core estimates using the classifier's own day-selection, and separately using an unsupervised clustering that never saw price data at all, confirms that the equity and credit results are not an artifact of exactly how the newsy-day threshold was drawn — those signs and rough magnitudes hold up across all three approaches. The VIX and the dollar are more sensitive to the choice of day-set and should be read with a little more caution than the rest of the table.

Figure 5. Three ways to draw the day-classification line, and one plain event-study check, mostly agree on sign



What the word list misses

The escalation/de-escalation language used to score news coverage is a fixed list built in advance, and a live, fast-moving conflict generates vocabulary no fixed list can anticipate — named agreements, facility names, invented shorthand. Checking the verified news text sentence by sentence against that list finds that 52.5% of sentences plainly about the war — describing strikes, blockades, casualties, and ceasefire terms — register zero matches. "Natanz nuclear enrichment complex" scores nothing because the list has no place-name knowledge at all; "rogue supertankers," a phrase this specific conflict's coverage invented, was never going to be on any pre-built list; the "memorandum-of-understanding ceasefire" names an agreement that did not exist before this war began.

This limitation sits at the sentence level and does not undermine the day-selection itself, which runs on aggregate coverage volume rather than phrase-matching — a spike in Iran-related reporting is detected whether or not any particular sentence uses recognized vocabulary. Where it does show up is in anything relying on the word list's read of a specific sentence's direction, which is part of why the word-list and a general-purpose language-model sentiment score agree only weakly at that fine-grained level even though they track each other reasonably well in aggregate. A language model reading the text directly, rather than counting phrases against a fixed list, would sidestep this — it does not need "Abqaiq" or "supertanker" pre-registered to understand what they mean in context — though building that as a full automated step was outside the scope of what could be tested and verified here.

Other frameworks worth knowing about

A handful of other established approaches could, in principle, answer a version of this question, and each was set aside for a specific reason rather than overlooked.

Approach	What it does	Why it wasn't used here
Geopolitical Risk Index (Caldara & Iacoviello, 2018)	Counts geopolitical-risk articles across ten major newspapers into a continuous index	Answers a related but different question — a risk <i>level</i> , not the variance shift this analysis is built around — and needs full-text access to ten specific papers that free data sources don't provide
Economic Policy Uncertainty Index (Baker, Bloom & Davis, 2016)	Same keyword-counting methodology, applied to policy uncertainty generally	Same limitation as above
Markov-switching volatility models (Hamilton, 1989)	Lets returns alone determine regime membership, with no news input at all	The market-only alternative already tested here (the unsupervised clustering) shows that a market-only day-classification is close to uninformative about Iran-specific risk by construction, for the same reason a return-only switching model would be
Structural VAR with sign restrictions (Uhlig, 2005)	Identifies the shock through theoretically motivated restrictions on how variables can respond, rather than through variance shifts	Requires specifying a full system and defensible restrictions across seventeen variables at once — a substantially larger undertaking that trades one set of assumptions for another rather than avoiding assumptions
Direct language-model scoring	Reads each day's text and outputs a calibrated severity score rather than counting fixed phrases	The natural fix for the vocabulary gap above; not built as a full pipeline here given the added cost and scope of running a model call per day across the sample

None of these are worse ideas — the Geopolitical Risk Index in particular is a well-established, Fed-published methodology. Each simply answers a different question, or needs data this analysis's free, single-session data sources could not supply.

A different war than 2003

The 2003 episode offers a natural benchmark, and one deliberate reason to expect a break from it: US crude output has roughly doubled and then some since then, from around 6 million barrels a day to around 14 million, which changes how a war-driven oil shock feeds through the domestic economy.

	2003 (Rigobon-Sack)	2026 (this analysis)	Consistent with the shift?
Oil	Rises	Rises (variance 6.2x higher on war days)	Yes
Credit spreads	Widen	Widen (high yield +4.8bp, investment grade +0.8bp)	Yes
Equities	Fall	Fall (S&P -0.65%, more abroad)	Yes
Gold	No significant response	No significant response	Yes
Treasury yields	Fall (-26bp on the ten-year)	Rise (+1.9bp on the ten-year)	No — as expected
Break-even inflation	Falls	Rises	No — as expected
Dollar	Falls	Rises	No — as expected

Every leg that was expected to hold steady with 2003 held; every leg expected to flip, flipped. The US-outperformance pattern in equities, and energy stocks moving with rather than against the broader market, is the same story told at the sector level: a much larger domestic oil industry turns a war-driven price spike from a uniformly negative shock into a genuinely mixed one for the US economy, which is consistent with yields rising on an inflation-and-growth read rather than falling on a pure flight-to-safety read the way they did when the 2003 shock was seen purely as a threat to demand.

Reading the caveats

A few limitations should travel with every number above. The core assumption behind the whole exercise — that nothing *other* than war risk was unusually volatile on exactly the eighteen flagged days — cannot be independently proven, only checked for plausibility, and the stability of the results when the day-count is widened or narrowed is reassuring on that front without being conclusive. The quiet comparison days are not perfectly free of war-related news in a conflict that ran six months, which biases the variance-share numbers in the earlier table downward, so they are best read as floors. "War risk" here is a single estimated factor standing in for several distinct things at once — the odds of further escalation, how long the conflict runs, how much of the Strait of Hormuz stays open — bundled together rather than separated out. The sample itself is small, eighteen days on each side, which leaves four of the seventeen coefficients (gold, Tel Aviv, defence stocks, and to a lesser extent the dollar) resting on weaker statistical footing than the rest and worth treating with real caution. And the eighteen event descriptions in the table above are a hand-verified research step rather than something the automated pipeline regenerates on its own — a genuine change in which days get flagged would need that verification redone.

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